



CHRIST
(DEEMED TO BE UNIVERSITY)
DELHI - NCR, INDIA

NLDL Programming

MCA-575

Assignment – 06

BY

HIMANSHU HEDA (24225013)

SUBMITTED TO

Dr. Manjula Shannhog

SCHOOL OF SCIENCES

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Importing the Libraries

```
import tensorflow
from tensorflow import keras
from tensorflow.keras import Sequential
from tensorflow.keras.layers import Dense, Flatten
from tensorflow.keras.datasets import mnist
```

```
import matplotlib.pyplot as plt
```

```
(x_train, y_train), (x_test, y_test) = mnist.load_data()
```

Downloading data from <https://storage.googleapis.com/tensorflow/tf-keras-datasets/mnist.npz>

11490434/11490434 [0m 32m
0m 37m 0m 1m9s 0m 1us/step

```
x_test
```

```
array([[0, 0, 0, ..., 0, 0, 0],
       [0, 0, 0, ..., 0, 0, 0],
       [0, 0, 0, ..., 0, 0, 0],
       ...,
       [0, 0, 0, ..., 0, 0, 0],
       [0, 0, 0, ..., 0, 0, 0],
       [0, 0, 0, ..., 0, 0, 0]],

       [[0, 0, 0, ..., 0, 0, 0],
       [0, 0, 0, ..., 0, 0, 0],
       [0, 0, 0, ..., 0, 0, 0],
       ...,
       [0, 0, 0, ..., 0, 0, 0],
       [0, 0, 0, ..., 0, 0, 0],
       [0, 0, 0, ..., 0, 0, 0]],

       [[0, 0, 0, ..., 0, 0, 0],
       [0, 0, 0, ..., 0, 0, 0],
       [0, 0, 0, ..., 0, 0, 0],
       ...,
       [0, 0, 0, ..., 0, 0, 0],
       [0, 0, 0, ..., 0, 0, 0],
       [0, 0, 0, ..., 0, 0, 0]],

       ...,

       [[0, 0, 0, ..., 0, 0, 0],
       [0, 0, 0, ..., 0, 0, 0],
       [0, 0, 0, ..., 0, 0, 0],
```

```

...,
[0, 0, 0, ..., 0, 0, 0],
[0, 0, 0, ..., 0, 0, 0],
[0, 0, 0, ..., 0, 0, 0]],

[[0, 0, 0, ..., 0, 0, 0],
[0, 0, 0, ..., 0, 0, 0],
[0, 0, 0, ..., 0, 0, 0],

...,
[0, 0, 0, ..., 0, 0, 0],
[0, 0, 0, ..., 0, 0, 0],
[0, 0, 0, ..., 0, 0, 0]],

[[0, 0, 0, ..., 0, 0, 0],
[0, 0, 0, ..., 0, 0, 0],
[0, 0, 0, ..., 0, 0, 0],

...,
[0, 0, 0, ..., 0, 0, 0],
[0, 0, 0, ..., 0, 0, 0],
[0, 0, 0, ..., 0, 0, 0]]], shape=(10000, 28, 28), dtype=uint8)

```

```
x_train
```

```

array([[[[0, 0, 0, ..., 0, 0, 0],
        [0, 0, 0, ..., 0, 0, 0],
        [0, 0, 0, ..., 0, 0, 0],

        ...,
        [0, 0, 0, ..., 0, 0, 0],
        [0, 0, 0, ..., 0, 0, 0],
        [0, 0, 0, ..., 0, 0, 0]],

        [[0, 0, 0, ..., 0, 0, 0],
        [0, 0, 0, ..., 0, 0, 0],
        [0, 0, 0, ..., 0, 0, 0],

        ...,
        [0, 0, 0, ..., 0, 0, 0],
        [0, 0, 0, ..., 0, 0, 0],
        [0, 0, 0, ..., 0, 0, 0]],

        [[0, 0, 0, ..., 0, 0, 0],
        [0, 0, 0, ..., 0, 0, 0],
        [0, 0, 0, ..., 0, 0, 0],

        ...,
        [0, 0, 0, ..., 0, 0, 0],
        [0, 0, 0, ..., 0, 0, 0],
        [0, 0, 0, ..., 0, 0, 0]],

        ...

```

```
...,
[[0, 0, 0, ..., 0, 0, 0],
 [0, 0, 0, ..., 0, 0, 0],
 [0, 0, 0, ..., 0, 0, 0],
 ...,
 [0, 0, 0, ..., 0, 0, 0],
 [0, 0, 0, ..., 0, 0, 0],
 [0, 0, 0, ..., 0, 0, 0]],
[[0, 0, 0, ..., 0, 0, 0],
 [0, 0, 0, ..., 0, 0, 0],
 [0, 0, 0, ..., 0, 0, 0],
 ...,
 [0, 0, 0, ..., 0, 0, 0],
 [0, 0, 0, ..., 0, 0, 0],
 [0, 0, 0, ..., 0, 0, 0]],
[[0, 0, 0, ..., 0, 0, 0],
 [0, 0, 0, ..., 0, 0, 0],
 [0, 0, 0, ..., 0, 0, 0],
 ...,
 [0, 0, 0, ..., 0, 0, 0],
 [0, 0, 0, ..., 0, 0, 0],
 [0, 0, 0, ..., 0, 0, 0]]], shape=(60000, 28, 28), dtype=uint8)
```

```
x_train.shape
```

(60000, 28, 28)

```
x_train[10]
```

```
array([[0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
        0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
        0, 0],
       [0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
        0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
        0, 0],
       [0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
        0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
        0, 0],
       [0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
        0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
        0, 0],
       [0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 42, 118,
        219, 166, 118, 118, 6, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0.]])
```

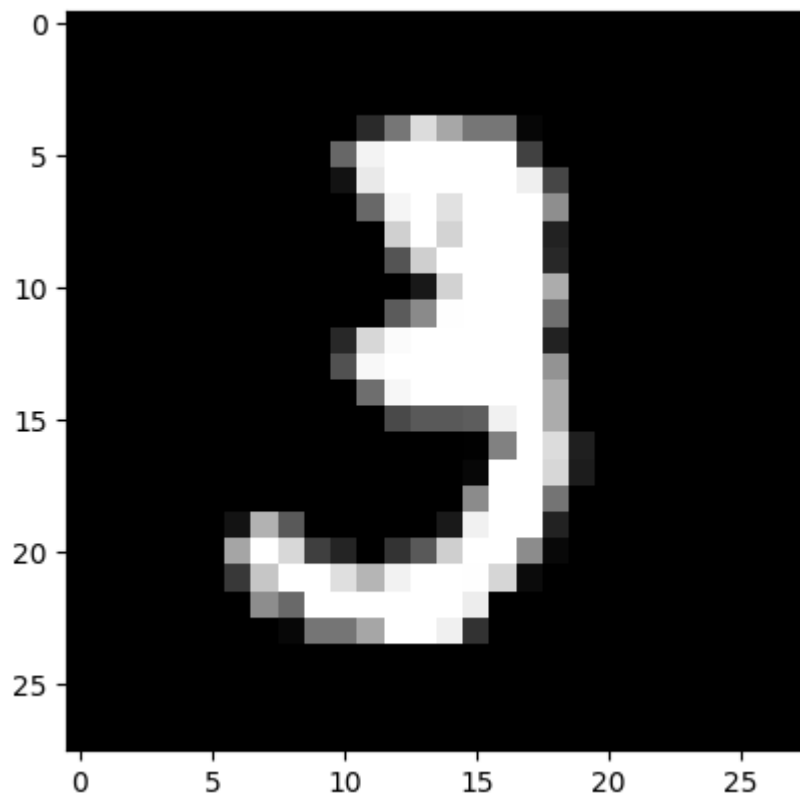
0, 0],
[0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 103, 242, 254,
254, 254, 254, 254, 66, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0],
[0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 18, 232, 254,
254, 254, 254, 254, 238, 70, 0, 0, 0, 0, 0, 0, 0,
0, 0],
[0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 104, 244,
254, 224, 254, 254, 254, 141, 0, 0, 0, 0, 0, 0, 0,
0, 0],
[0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 207,
254, 210, 254, 254, 254, 34, 0, 0, 0, 0, 0, 0, 0,
0, 0],
[0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 84,
206, 254, 254, 254, 254, 41, 0, 0, 0, 0, 0, 0, 0,
0, 0],
[0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
24, 209, 254, 254, 254, 171, 0, 0, 0, 0, 0, 0, 0,
0, 0],
[0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 91,
137, 253, 254, 254, 254, 112, 0, 0, 0, 0, 0, 0, 0,
0, 0],
[0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 40, 214, 250,
254, 254, 254, 254, 254, 34, 0, 0, 0, 0, 0, 0, 0,
0, 0],
[0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 81, 247, 254,
254, 254, 254, 254, 146, 0, 0, 0, 0, 0, 0, 0,
0, 0],
[0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 110, 246,
254, 254, 254, 254, 254, 171, 0, 0, 0, 0, 0, 0, 0,
0, 0],
[0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 73,
89, 89, 93, 240, 254, 171, 0, 0, 0, 0, 0, 0, 0,
0, 0],
[0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0, 1, 128, 254, 219, 31, 0, 0, 0, 0, 0, 0, 0,
0, 0],
[0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0, 7, 254, 254, 214, 28, 0, 0, 0, 0, 0, 0, 0,
0, 0],
[0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0, 138, 254, 254, 116, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0],
[0, 0, 0, 0, 0, 0, 19, 177, 90, 0, 0, 0, 0, 0,
0, 25, 240, 254, 254, 34, 0, 0, 0, 0, 0, 0, 0,
0, 0],

```
[ 0, 0, 0, 0, 0, 0, 164, 254, 215, 63, 36, 0, 51,
 89, 206, 254, 254, 139, 8, 0, 0, 0, 0, 0, 0, 0,
 0, 0],
[ 0, 0, 0, 0, 0, 0, 57, 197, 254, 254, 222, 180, 241,
254, 254, 253, 213, 11, 0, 0, 0, 0, 0, 0, 0, 0,
 0, 0],
[ 0, 0, 0, 0, 0, 0, 0, 140, 105, 254, 254, 254, 254,
254, 254, 236, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
 0, 0],
[ 0, 0, 0, 0, 0, 0, 0, 0, 7, 117, 117, 165, 254,
254, 239, 50, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
 0, 0],
[ 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
 0, 0],
[ 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
 0, 0],
[ 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
 0, 0]], dtype=uint8)
```

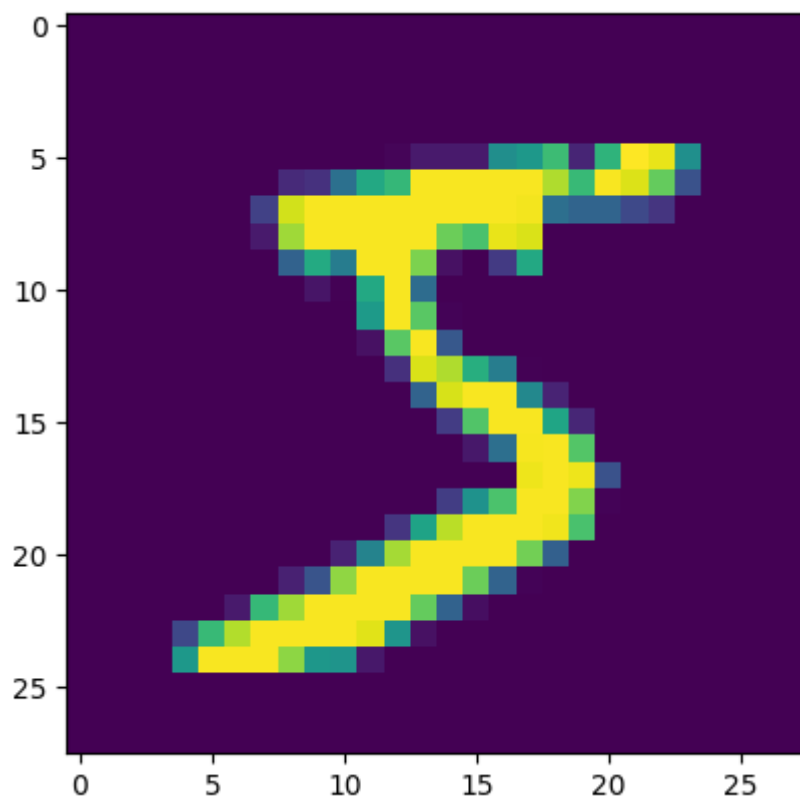
```
y_train
```

```
array([5, 0, 4, ..., 5, 6, 8], shape=(60000,), dtype=uint8)
```

```
plt.imshow(x_train[10], cmap='gray')
```



```
plt.imshow(x_train[0])
```



```
plt.imshow(x_train[1])
```


[0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0, 0, 38, 48, 48, 22, 0, 0, 0, 0, 0, 0, 0,
0, 0],

[0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
62, 97, 198, 243, 254, 254, 212, 27, 0, 0, 0, 0, 0, 0,
0, 0],

[0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 67,
172, 254, 254, 225, 218, 218, 237, 248, 40, 0, 21, 164, 187,
0, 0],

[0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 89, 219,
254, 97, 67, 14, 0, 0, 92, 231, 122, 23, 203, 236, 59,
0, 0],

[0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 25, 217, 242,
92, 4, 0, 0, 0, 0, 4, 147, 253, 240, 232, 92, 0,
0, 0],

[0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 101, 255, 92,
0, 0, 0, 0, 0, 0, 105, 254, 254, 177, 11, 0, 0, 0,
0, 0],

[0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 167, 244, 41,
0, 0, 0, 7, 76, 199, 238, 239, 94, 10, 0, 0, 0, 0,
0, 0],

[0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 192, 121, 0,
0, 2, 63, 180, 254, 233, 126, 0, 0, 0, 0, 0, 0, 0,
0, 0],

[0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 190, 196, 14,
2, 97, 254, 252, 146, 52, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0],

[0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 130, 225, 71,
180, 232, 181, 60, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0],

[0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 130, 254, 254,
230, 46, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0],

[0, 0, 0, 0, 0, 0, 0, 0, 0, 6, 77, 244, 254, 162,
4, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0],

[0, 0, 0, 0, 0, 0, 0, 0, 0, 110, 254, 218, 254, 116,
0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0],

[0, 0, 0, 0, 0, 0, 0, 131, 254, 154, 28, 213, 86,
0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0],

[0, 0, 0, 0, 0, 0, 66, 209, 153, 19, 19, 233, 60,
0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0],

[0, 0, 0, 0, 0, 0, 142, 254, 165, 0, 14, 216, 167,

```

    0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
    0, 0],
[ 0, 0, 0, 0, 0, 0, 0, 90, 254, 175, 0, 18, 229, 92,
 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
 0, 0],
[ 0, 0, 0, 0, 0, 0, 0, 26, 229, 249, 176, 222, 244, 44,
 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
 0, 0],
[ 0, 0, 0, 0, 0, 0, 0, 0, 73, 193, 197, 134, 0, 0,
 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
 0, 0],
[ 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
 0, 0],
[ 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
 0, 0],
[ 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
 0, 0]], dtype=uint8)

```

```
y_train[59999]
```

```
np.uint8(8)
```

```
x_train
```

```

array([[[[0, 0, 0, ..., 0, 0, 0],
         [0, 0, 0, ..., 0, 0, 0],
         [0, 0, 0, ..., 0, 0, 0],
         ...,
         [0, 0, 0, ..., 0, 0, 0],
         [0, 0, 0, ..., 0, 0, 0],
         [0, 0, 0, ..., 0, 0, 0]],

        [[0, 0, 0, ..., 0, 0, 0],
         [0, 0, 0, ..., 0, 0, 0],
         [0, 0, 0, ..., 0, 0, 0],
         ...,
         [0, 0, 0, ..., 0, 0, 0],
         [0, 0, 0, ..., 0, 0, 0],
         [0, 0, 0, ..., 0, 0, 0]],

        [[0, 0, 0, ..., 0, 0, 0],
         [0, 0, 0, ..., 0, 0, 0],
         [0, 0, 0, ..., 0, 0, 0],
         ...,
         [0, 0, 0, ..., 0, 0, 0],
         [0, 0, 0, ..., 0, 0, 0],
         [0, 0, 0, ..., 0, 0, 0]]],
      dtype=uint8)

```

$$\begin{bmatrix} 0, 0, 0, \dots, 0, 0, 0, \\ 0, 0, 0, \dots, 0, 0, 0, \\ 0, 0, 0, \dots, 0, 0, 0 \end{bmatrix},$$
$$\begin{bmatrix} 0 & 0 & 0 & \dots & 0 & 0 & 0 \\ 0 & 0 & 0 & \dots & 0 & 0 & 0 \\ 0 & 0 & 0 & \dots & 0 & 0 & 0 \end{bmatrix}$$
$$\begin{bmatrix} 0, 0, 0, \dots, 0, 0, 0, \\ 0, 0, 0, \dots, 0, 0, 0, \\ 0, 0, 0, \dots, 0, 0, 0 \end{bmatrix},$$
$$\begin{bmatrix} 0 & 0 & 0 & \dots & 0 & 0 & 0 \\ 0 & 0 & 0 & \dots & 0 & 0 & 0 \\ 0 & 0 & 0 & \dots & 0 & 0 & 0 \end{bmatrix}$$
$$\begin{bmatrix} 0, 0, 0, \dots, 0, 0, 0, \\ 0, 0, 0, \dots, 0, 0, 0, \\ 0, 0, 0, \dots, 0, 0, 0 \end{bmatrix},$$
$$\begin{bmatrix} 0 & 0 & 0 & \dots & 0 & 0 & 0 \\ 0 & 0 & 0 & \dots & 0 & 0 & 0 \\ 0 & 0 & 0 & \dots & 0 & 0 & 0 \end{bmatrix}$$
$$\begin{bmatrix} 0, 0, 0, \dots, 0, 0, 0, \\ 0, 0, 0, \dots, 0, 0, 0, \\ 0, 0, 0, \dots, 0, 0, 0 \end{bmatrix}]$$

(10000, 28, 28)

```
x_train[0]
```

```
array([[0., 0., 0., 0., 0.,
        0., 0., 0., 0., 0.,
        0., 0., 0., 0., 0.,
        0., 0., 0., 0., 0.,
        0., 0., 0.],
       [0., 0., 0., 0., 0.,
        0., 0., 0., 0., 0.,
        0., 0., 0., 0., 0.]])
```

0. , 0. , 0. , 0. , 0. ,
 0. , 0. , 0. , 0. , 0. ,
 0. , 0. , 0.],
 [0. , 0. , 0. , 0. , 0. ,
 0. , 0. , 0. , 0. , 0. ,
 0. , 0. , 0. , 0. , 0. ,
 0. , 0. , 0. , 0. , 0. ,
 0. , 0. , 0. , 0. , 0. ,
 0. , 0. , 0.],
 [0. , 0. , 0. , 0. , 0. ,
 0. , 0. , 0. , 0. , 0. ,
 0. , 0. , 0. , 0. , 0. ,
 0. , 0. , 0. , 0. , 0. ,
 0. , 0. , 0. , 0. , 0. ,
 0. , 0. , 0.],
 [0. , 0. , 0. , 0. , 0. ,
 0. , 0. , 0. , 0. , 0. ,
 0. , 0. , 0.01176471, 0.07058824, 0.07058824,
 0.07058824, 0.49411765, 0.53333333, 0.68627451, 0.10196078,
 0.65098039, 1. , 0.96862745, 0.49803922, 0. ,
 0. , 0. , 0.],
 [0. , 0. , 0. , 0. , 0. ,
 0. , 0. , 0. , 0.11764706, 0.14117647,
 0.36862745, 0.60392157, 0.66666667, 0.99215686, 0.99215686,
 0.99215686, 0.99215686, 0.99215686, 0.88235294, 0.6745098 ,
 0.99215686, 0.94901961, 0.76470588, 0.25098039, 0. ,
 0. , 0. , 0.],
 [0. , 0. , 0. , 0. , 0. ,
 0. , 0. , 0.19215686, 0.93333333, 0.99215686,
 0.99215686, 0.99215686, 0.99215686, 0.99215686, 0.99215686,
 0.99215686, 0.99215686, 0.98431373, 0.36470588, 0.32156863,
 0.32156863, 0.21960784, 0.15294118, 0. , 0. ,
 0. , 0. , 0.],
 [0. , 0. , 0. , 0. , 0. ,
 0. , 0. , 0.07058824, 0.85882353, 0.99215686,
 0.99215686, 0.99215686, 0.99215686, 0.99215686, 0.77647059,
 0.71372549, 0.96862745, 0.94509804, 0. , 0. ,
 0. , 0. , 0. , 0. , 0. ,
 0. , 0. , 0.],
 [0. , 0. , 0. , 0. , 0. ,

0. , 0. , 0. , 0.31372549, 0.61176471,
0.41960784, 0.99215686, 0.99215686, 0.80392157, 0.04313725,
0. , 0.16862745, 0.60392157, 0. , 0. ,
0. , 0. , 0. , 0. , 0. ,
0. , 0. , 0.],
[0. , 0. , 0. , 0. , 0. ,
0. , 0. , 0. , 0. , 0.05490196,
0.00392157, 0.60392157, 0.99215686, 0.35294118, 0. ,
0. , 0. , 0. , 0. , 0. ,
0. , 0. , 0. , 0. , 0. ,
0. , 0. , 0.],
[0. , 0. , 0. , 0. , 0. ,
0. , 0. , 0. , 0. , 0. ,
0. , 0.54509804, 0.99215686, 0.74509804, 0.00784314,
0. , 0. , 0. , 0. , 0. ,
0. , 0. , 0. , 0. , 0. ,
0. , 0. , 0.],
[0. , 0. , 0. , 0. , 0. ,
0. , 0. , 0. , 0. , 0. ,
0. , 0.04313725, 0.74509804, 0.99215686, 0.2745098 ,
0. , 0. , 0. , 0. , 0. ,
0. , 0. , 0. , 0. , 0. ,
0. , 0. , 0.],
[0. , 0. , 0. , 0. , 0. ,
0. , 0. , 0. , 0. , 0. ,
0. , 0. , 0.1372549 , 0.94509804, 0.88235294,
0.62745098, 0.42352941, 0.00392157, 0. , 0. ,
0. , 0. , 0. , 0. , 0. ,
0. , 0. , 0.],
[0. , 0. , 0. , 0. , 0. ,
0. , 0. , 0. , 0. , 0. ,
0. , 0. , 0. , 0.31764706, 0.94117647,
0.99215686, 0.99215686, 0.46666667, 0.09803922, 0. ,
0. , 0. , 0. , 0. , 0. ,
0. , 0. , 0.],
[0. , 0. , 0. , 0. , 0. ,
0. , 0. , 0. , 0. , 0. ,
0. , 0. , 0. , 0. , 0.17647059,
0.72941176, 0.99215686, 0.99215686, 0.58823529, 0.10588235,
0. , 0. , 0. , 0. , 0. ,
0. , 0. , 0.],
[0. , 0. , 0. , 0. , 0. ,
0. , 0. , 0. , 0. , 0. ,
0. , 0. , 0. , 0. , 0. ,
0.0627451 , 0.36470588, 0.98823529, 0.99215686, 0.73333333,
0. , 0. , 0. , 0. , 0. ,

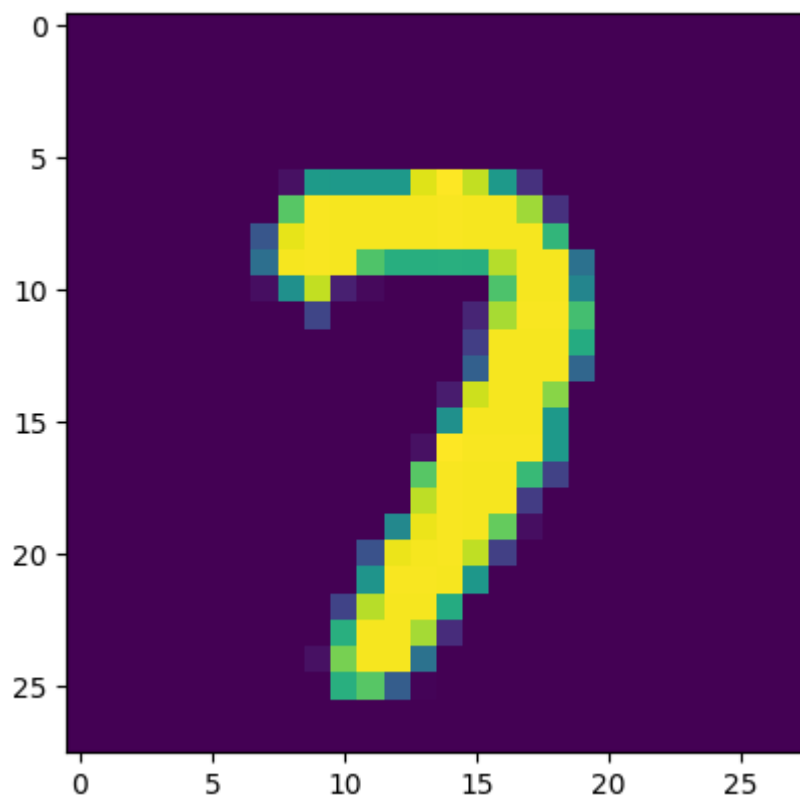
0. , 0. , 0.],
 [0. , 0. , 0. , 0. , 0. ,
 0. , 0. , 0. , 0. , 0. ,
 0. , 0. , 0. , 0. , 0. ,
 0. , 0. , 0.97647059, 0.99215686, 0.97647059,
 0.25098039, 0. , 0. , 0. , 0. ,
 0. , 0. , 0.],
 [0. , 0. , 0. , 0. , 0. ,
 0. , 0. , 0. , 0. , 0. ,
 0. , 0. , 0. , 0. , 0.18039216,
 0.50980392, 0.71764706, 0.99215686, 0.99215686, 0.81176471,
 0.00784314, 0. , 0. , 0. , 0. ,
 0. , 0. , 0.],
 [0. , 0. , 0. , 0. , 0. ,
 0. , 0. , 0. , 0. , 0. ,
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 0. , 0. , 0. , 0. , 0. ,
 0. , 0. , 0.],
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 0.51764706, 0.0627451 , 0. , 0. , 0. ,

```

0.    ,0.    ,0.    ,0.    ,0.    ,
0.    ,0.    ,0.    ,0.    ,0.    ,
0.    ,0.    ,0.    ],
[0.    ,0.    ,0.    ,0.    ,0.    ,
0.    ,0.    ,0.    ,0.    ,0.    ,
0.    ,0.    ,0.    ,0.    ,0.    ,
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0.    ,0.    ,0.    ],
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0.    ,0.    ,0.    ],
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0.    ,0.    ,0.    ,0.    ,0.    ,
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0.    ,0.    ,0.    ,0.    ,0.    ,
0.    ,0.    ,0.    ,0.    ,0.    ,
0.    ,0.    ,0.    ]]

```

```
plt.imshow(x_train[1300])
```



```
model = Sequential()
```

```
model.add(Flatten(input_shape=(28, 28)))  
model.add(Dense(128, activation='relu'))  
model.add(Dense(10, activation='softmax'))
```

c:\Users\himan\AppData\Local\Programs\Python\Python313\Lib\site-packages\keras\src\layers\reshaping\flatten.py:37: UserWarning: Do not pass an `input\_shape`/`input\_dim` argument to a layer. When using Sequential models, prefer using an `Input(shape)` object as the first layer in the model instead.

```
super().__init__(**kwargs)
```

```
model.summary()
```

Model: "sequential"

Layer (type)	Output Shape	Param #
flatten (Flatten)	(None, 784)	0
dense (Dense)	(None, 128)	100,480
dense_1 (Dense)	(None, 10)	1,290

Total params: 101,770 (397.54 KB)

Trainable params: 101,770 (397.54 KB)

Non-trainable params: 0 (0.00 B)

```
model.compile(loss='sparse_categorical_crossentropy', optimizer='adam',  
metrics=['accuracy'])
```

```
hist=model.fit(x_train, y_train, epochs=10, validation_split=0.2)
```

Epoch 1/10

1m1500/1500 0m 32m
0m 37m 0m 1m9s 0m 5ms/step - accuracy: 0.9187 - loss: 0.2849 - val_accuracy:
0.9582 - val_loss: 0.1491

Epoch 2/10

1m1500/1500 0m 32m
0m 37m 0m 1m11s 0m 7ms/step - accuracy: 0.9632 - loss: 0.1240 - val_accuracy:
0.9678 - val_loss: 0.1139

Epoch 3/10

1m1500/1500 0m 32m
0m 37m 0m 1m11s 0m 7ms/step - accuracy: 0.9745 - loss: 0.0852 - val_accuracy:
0.9675 - val_loss: 0.1087

Epoch 4/10

1m1500/1500 0m 32m
0m 37m 0m 1m10s 0m 7ms/step - accuracy: 0.9811 - loss: 0.0638 - val_accuracy:
0.9722 - val_loss: 0.0934

Epoch 5/10

□[1m1500/1500□[0m □[32m_____

 □[0m□[37m□[0m □[1m10s□[0m 7ms/step - accuracy: 0.9843 - loss: 0.0504 - val_accuracy: 0.9709 - val_loss: 0.0967

 Epoch 6/10

 □[1m1500/1500□[0m □[32m_____

 □[0m□[37m□[0m □[1m10s□[0m 7ms/step - accuracy: 0.9884 - loss: 0.0384 - val_accuracy: 0.9742 - val_loss: 0.0883

 Epoch 7/10

 □[1m1500/1500□[0m □[32m_____

 □[0m□[37m□[0m □[1m10s□[0m 7ms/step - accuracy: 0.9908 - loss: 0.0300 - val_accuracy: 0.9776 - val_loss: 0.0859

 Epoch 8/10

 □[1m1500/1500□[0m □[32m_____

 □[0m□[37m□[0m □[1m11s□[0m 7ms/step - accuracy: 0.9926 - loss: 0.0245 - val_accuracy: 0.9712 - val_loss: 0.1079

 Epoch 9/10

 □[1m1500/1500□[0m □[32m_____

 □[0m□[37m□[0m □[1m10s□[0m 7ms/step - accuracy: 0.9945 - loss: 0.0196 - val_accuracy: 0.9746 - val_loss: 0.0984

 Epoch 10/10

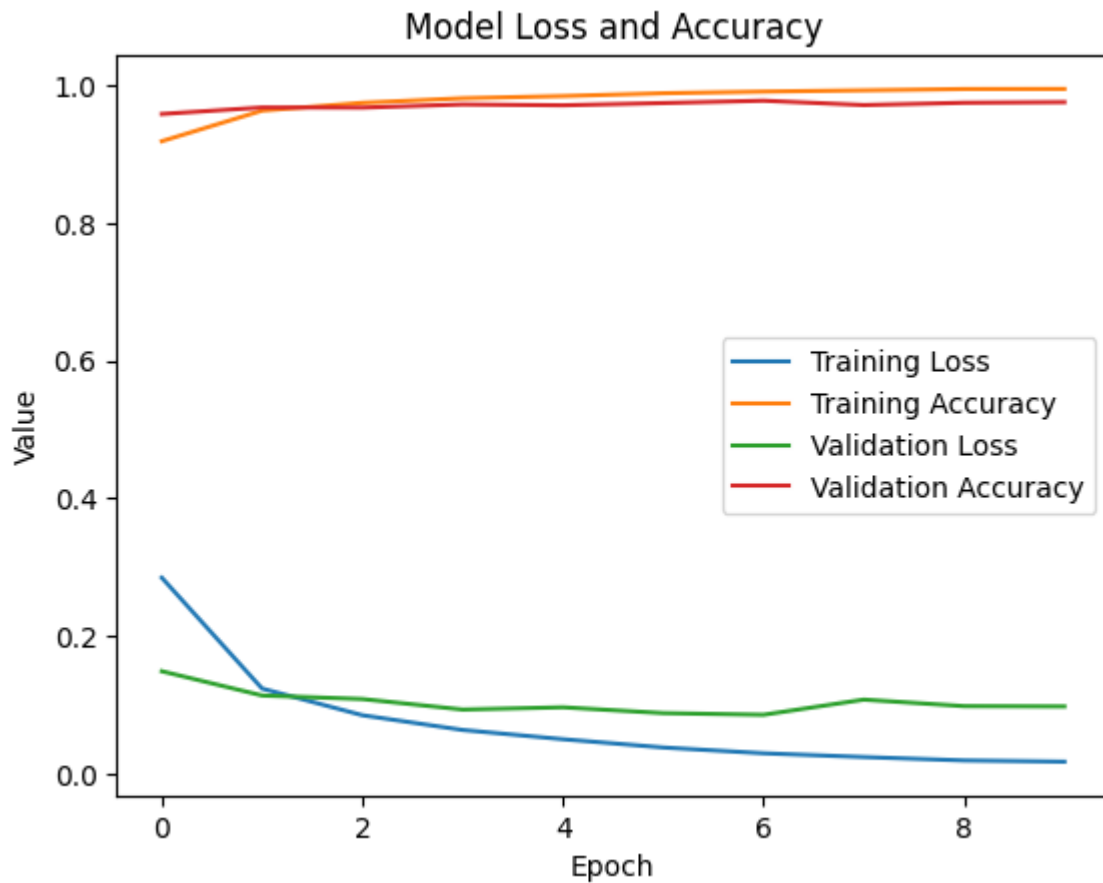
 □[1m1500/1500□[0m □[32m_____

 □[0m□[37m□[0m □[1m11s□[0m 7ms/step - accuracy: 0.9946 - loss: 0.0178 - val_accuracy: 0.9755 - val_loss: 0.0980

```

plt.plot(hist.history['loss'], label='Training Loss')
plt.plot(hist.history['accuracy'], label='Training Accuracy')
plt.plot(hist.history['val_loss'], label='Validation Loss')
plt.plot(hist.history['val_accuracy'], label='Validation Accuracy')
plt.xlabel('Epoch')
plt.ylabel('Value')
plt.title('Model Loss and Accuracy')
plt.legend()
plt.show()

```



```
model.predict(x_test)
```

```
[[1m313/313][0m][32m-----
```

```
[0m][37m[0m][1mls[0m 2ms/step
```

```
array([[0., 0., 0., ..., 1., 0., 0.],
       [0., 0., 1., ..., 0., 0., 0.],
       [0., 1., 0., ..., 0., 0., 0.],
       ...,
       [0., 0., 0., ..., 0., 0., 0.],
       [0., 0., 0., ..., 0., 0., 0.],
       [0., 0., 0., ..., 0., 0., 0.]], shape=(10000, 10), dtype=float32)
```

```
y_prob = model.predict(x_test)
```

```
[1m313/313][0m][32m-----
```

```
[0m][37m[0m][1mls[0m 2ms/step
```

```
y_prob.argmax(axis=1)
```

```
array([7, 2, 1, ..., 4, 5, 6], shape=(10000,))
```

```
y_pred = y_prob.argmax(axis=1)
```

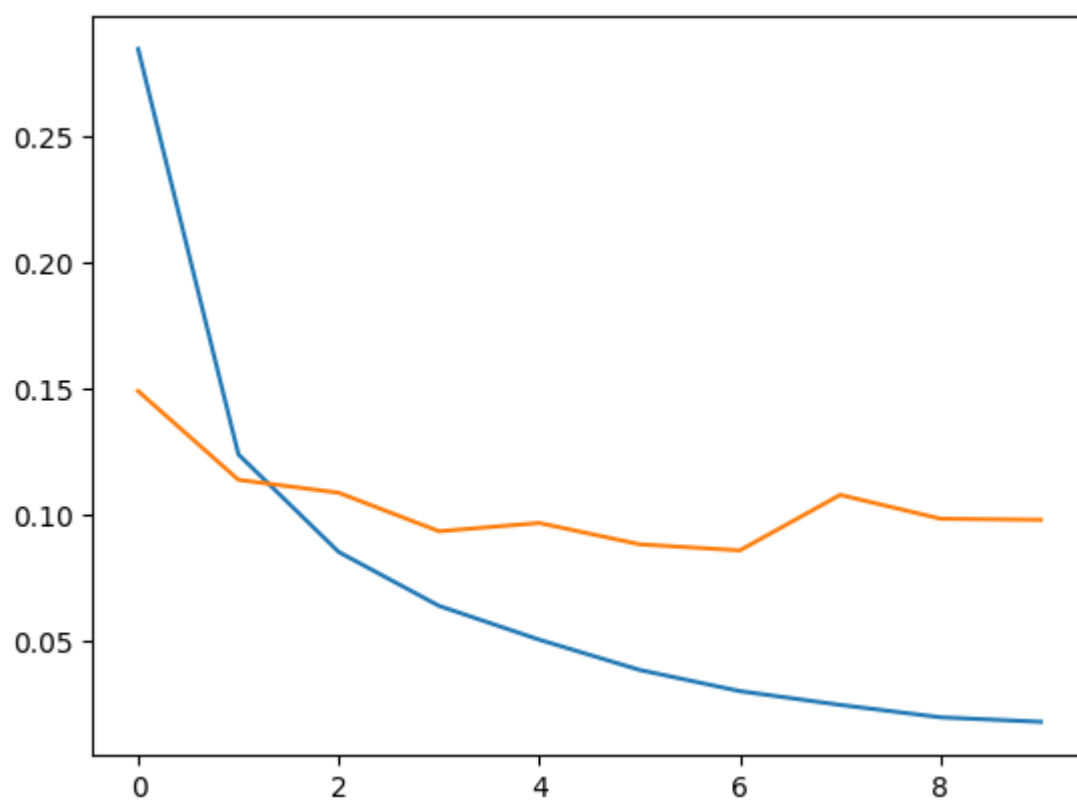
```
from sklearn.metrics import accuracy_score
```

```
acc = accuracy_score(y_test, y_pred)
```

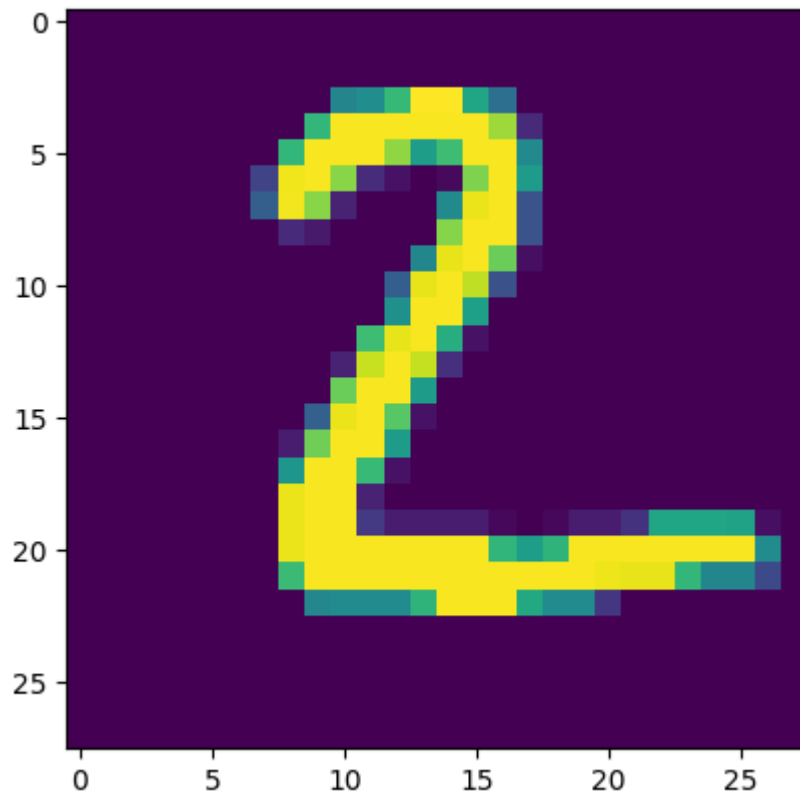
```
acc
```

0.9748

```
plt.plot(hist.history['loss'])  
plt.plot(hist.history['val_loss'])
```



```
plt.imshow(x_test[1])
```



```
y_pred[1]
```

```
np.int64(2)
```

```
model.predict(x_test[1].reshape(1, 28, 28))
```

□[1m1/1□[0m □[32m—————□[0m□[37m□[0m
 □[1m0s□[0m 45ms/step
 array([[0., 0., 1., 0., 0., 0., 0., 0., 0., 0.], dtype=float32)

```
model.predict(x_test[1].reshape(1, 28, 28)).argmax(axis=1)
```

□[1m1/1□[0m □[32m—————□[0m□[37m□[0m
 □[1m0s□[0m 26ms/step

```
array([2])
```